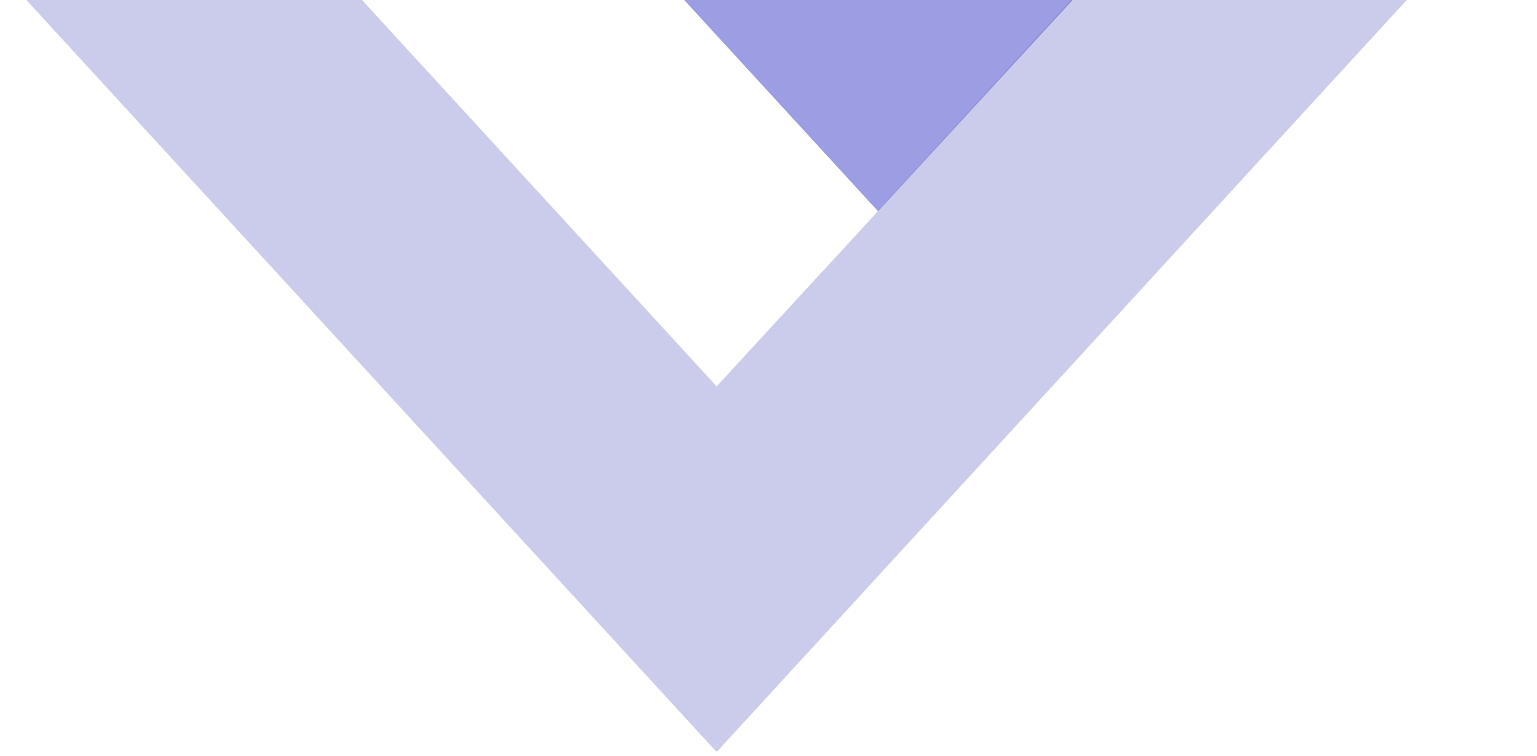




PREDICTIVE MODEL FOR DSC DONOR'S RE-ACTIVATION CAMPAIGN

Aarsi Poddar
Eswar Karthikeyan
Selima Medhioub

INTRODUCTION



- Non-profits often struggle to re-engage inactive donors.
- DSC aims to improve its reactivation campaign by focusing only on donors most likely to respond.
- A predictive model can help target the right people, reduce costs, and increase campaign impact.

TARGET DEFINITION



Goal: Re-activate inactive donors through targeted mail campaigns



Campaign Economics: €0.80 per letter cost

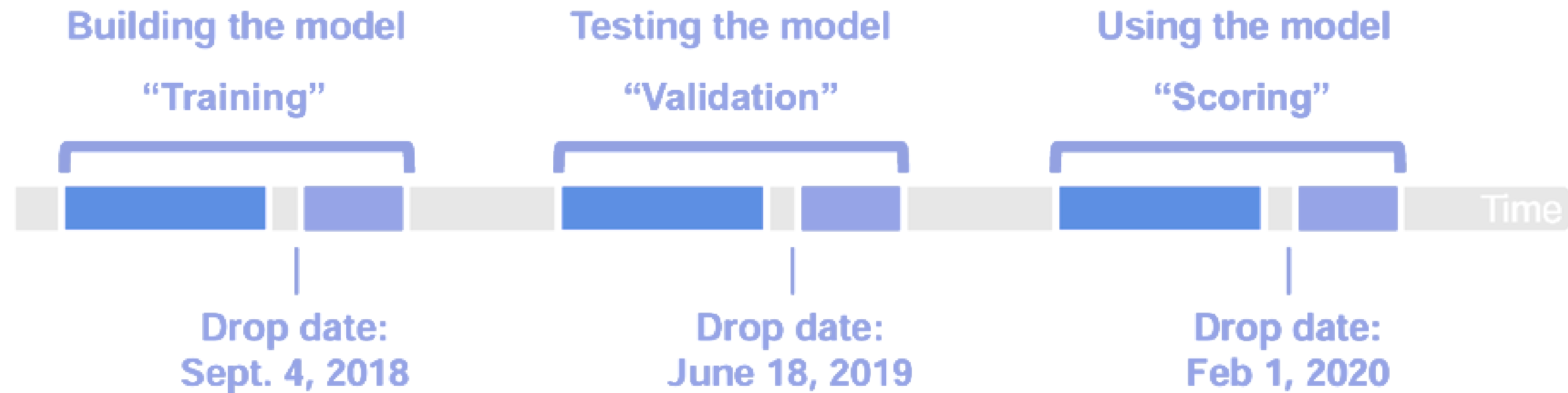


Focus: Donors contributing $\geq$ €30 for meaningful impact



Approach: Random selection with 1.8% baseline response rate

TIMELINE



- **1900 days back** = 5+ years of donor history (recency, frequency, amounts)
- **1 day gap** = Campaign date (decision point)
- **250 days forward** = Prediction window (did they donate $\geq$ €30?)

XGBOOST: OUR CHOSEN MODEL

Gradient Boosting

0.65

AUC Score

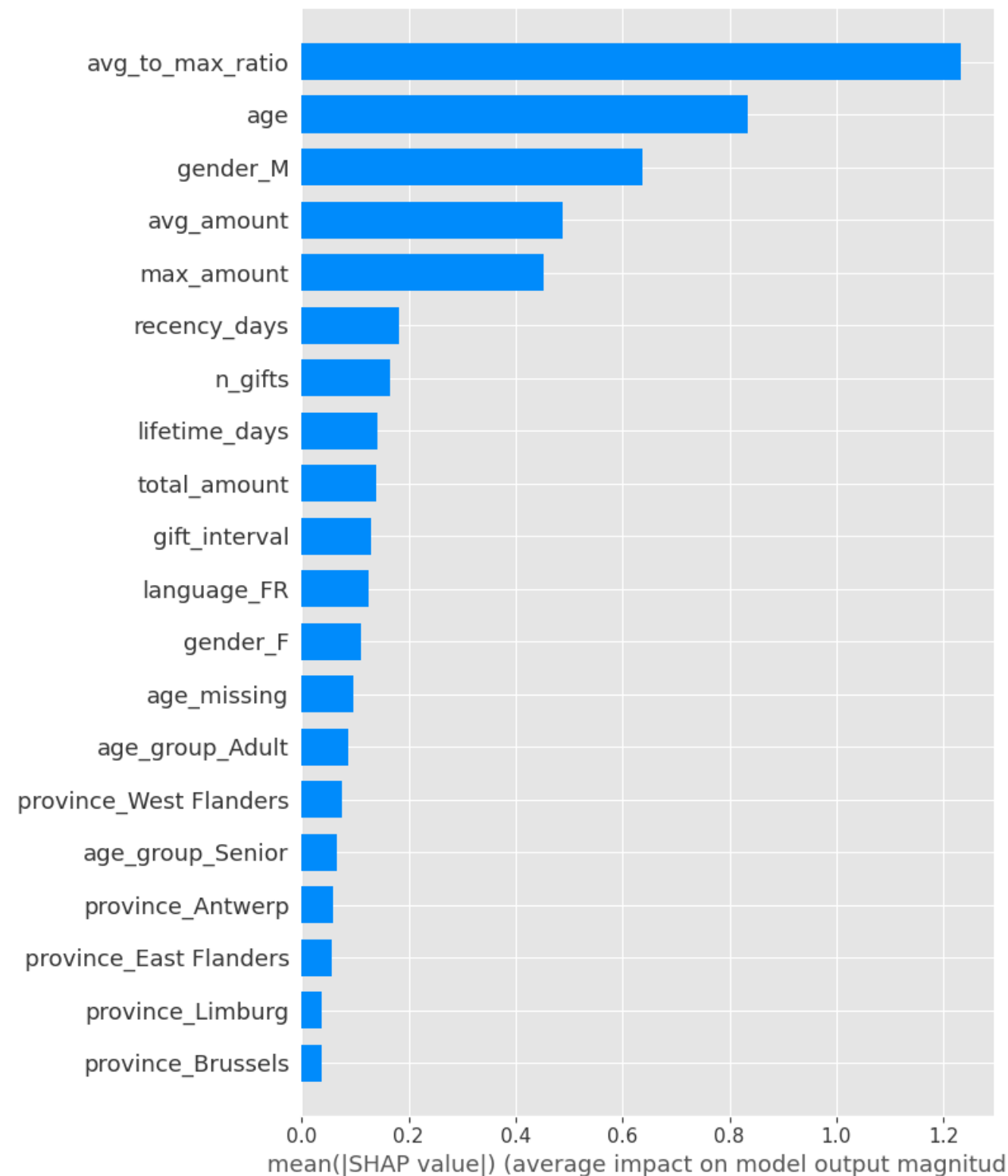
XGBoost ✓

0.80

AUC Score

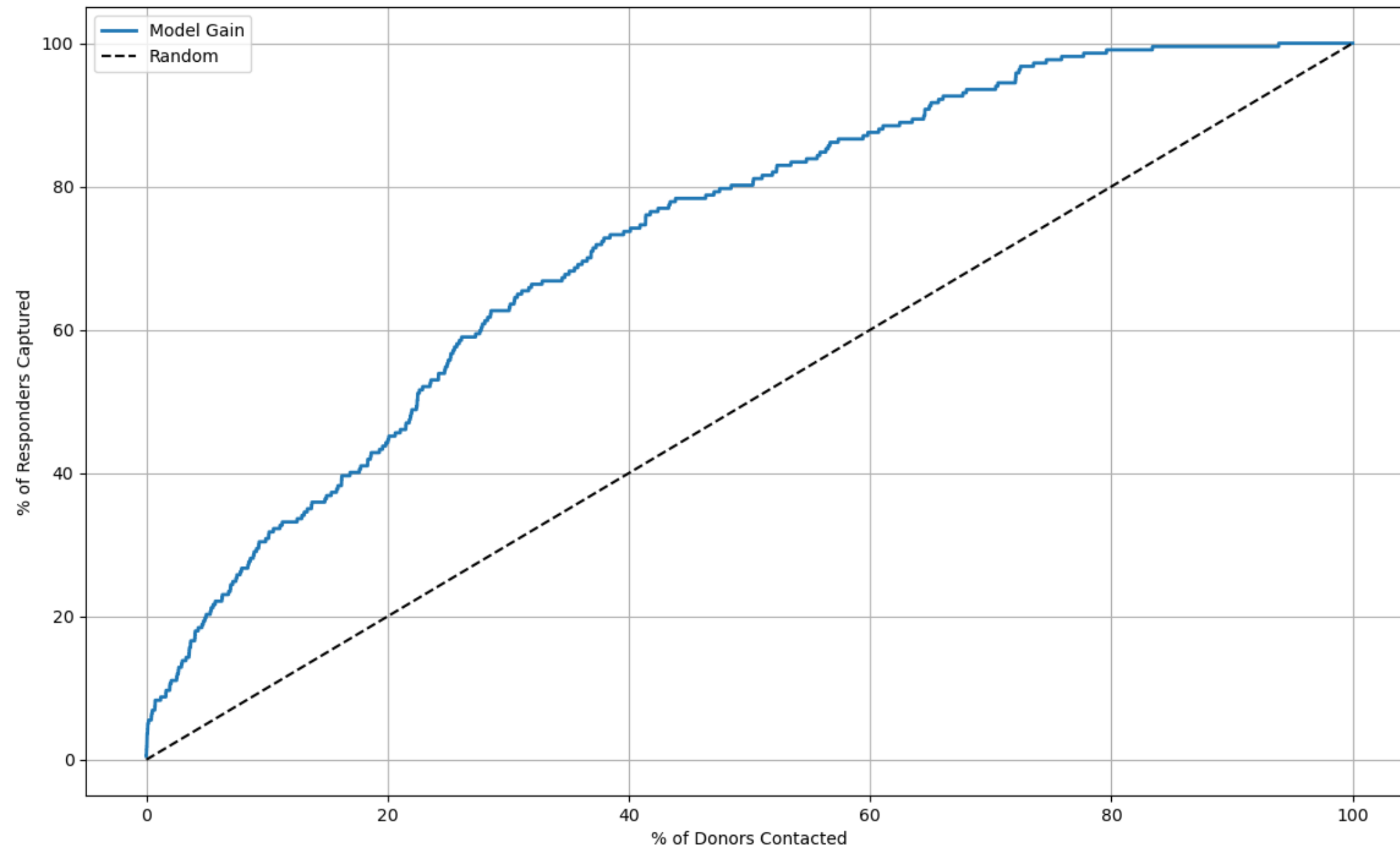
23% Better Performance

KEY PREDICTION DRIVERS



- Donation consistency
- Age
- Gender
- Average donation amount by donor
- Maximum donation by donor

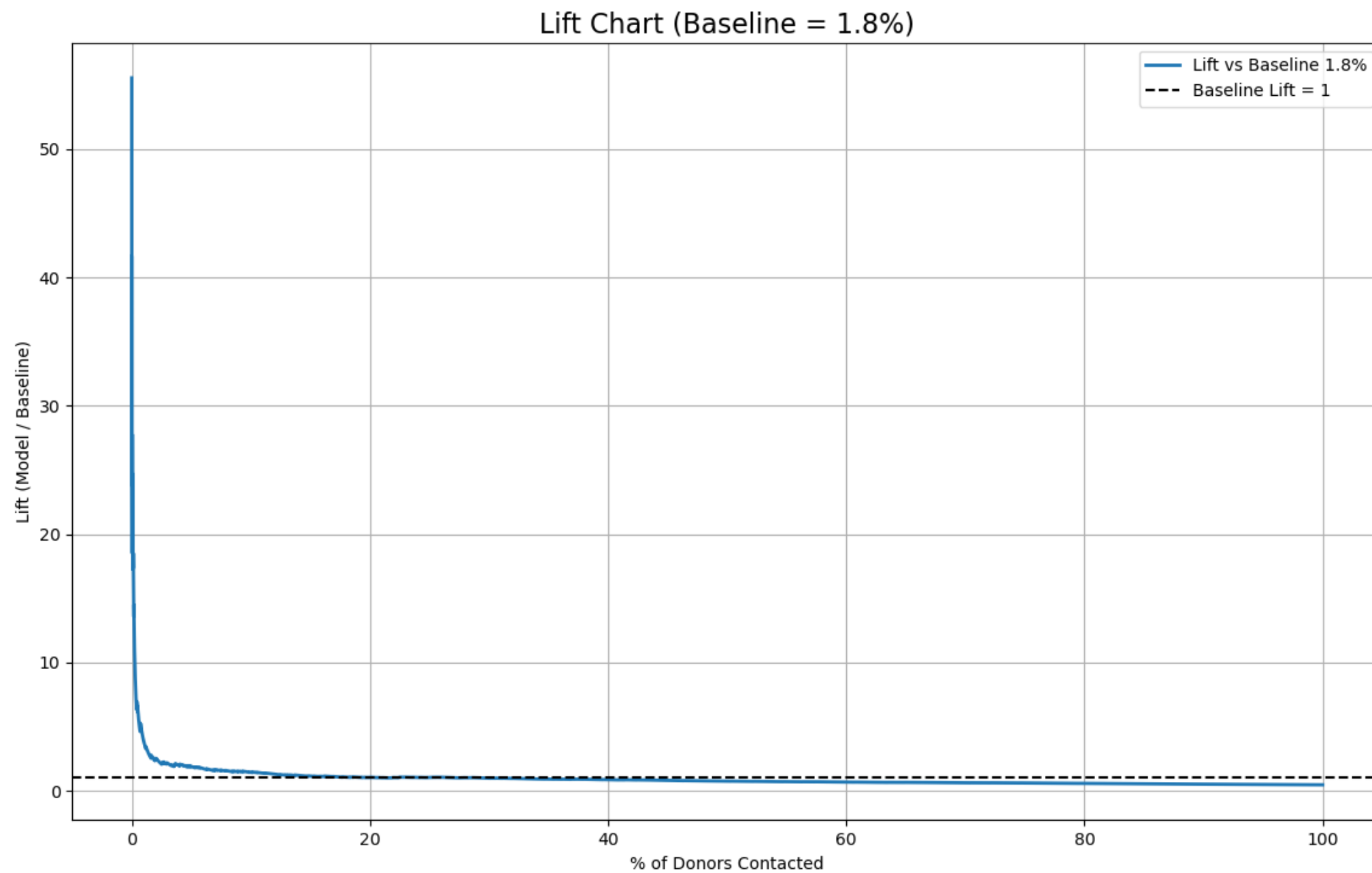
CUMULATIVE GAIN ANALYSIS



Top 10% captures – 35–40% of responders

Much higher than random (10%)

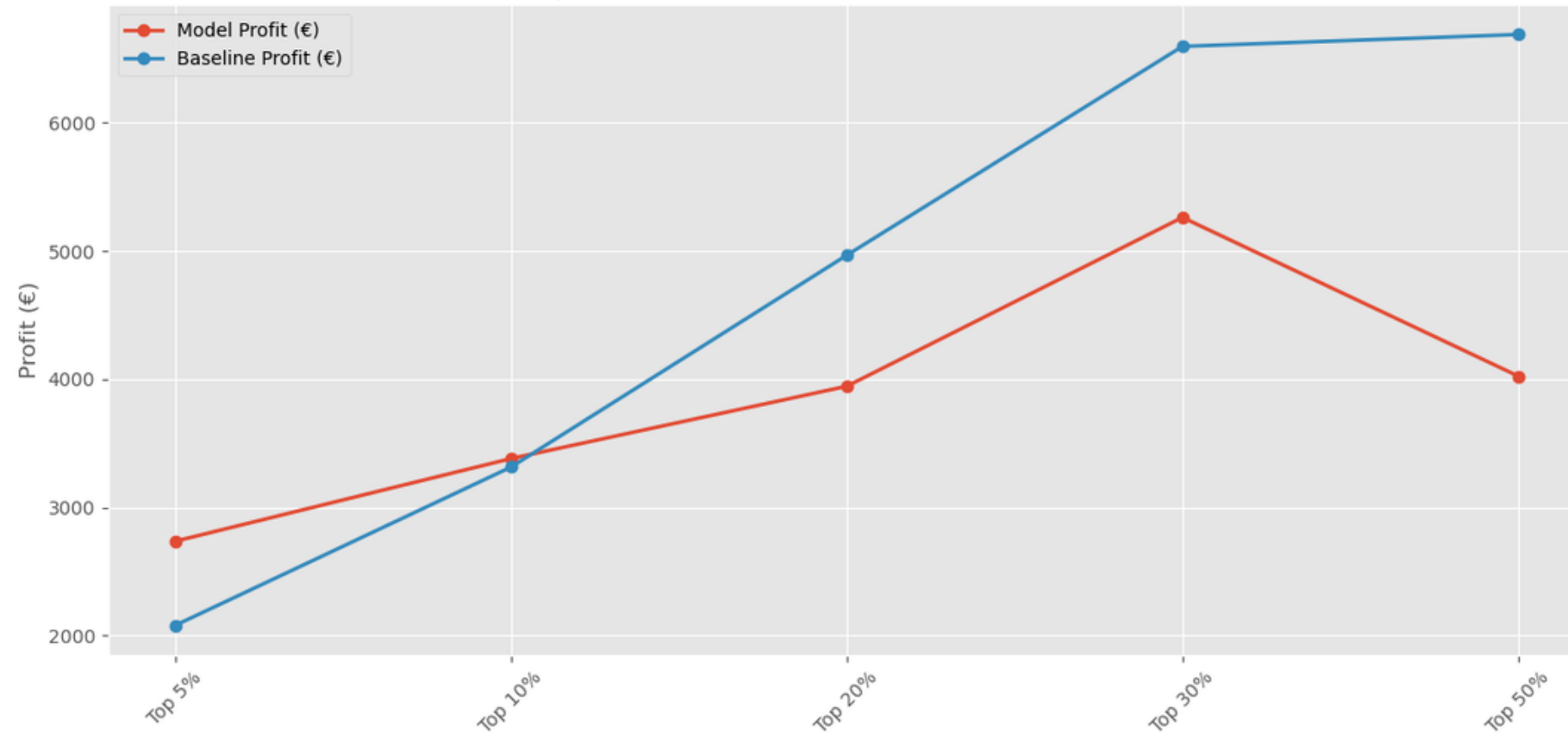
LIFT OVER BASELINE



Top 10% donors
1.46×
more likely to respond than
baseline response rate

PROFIT & ROI ANALYSIS

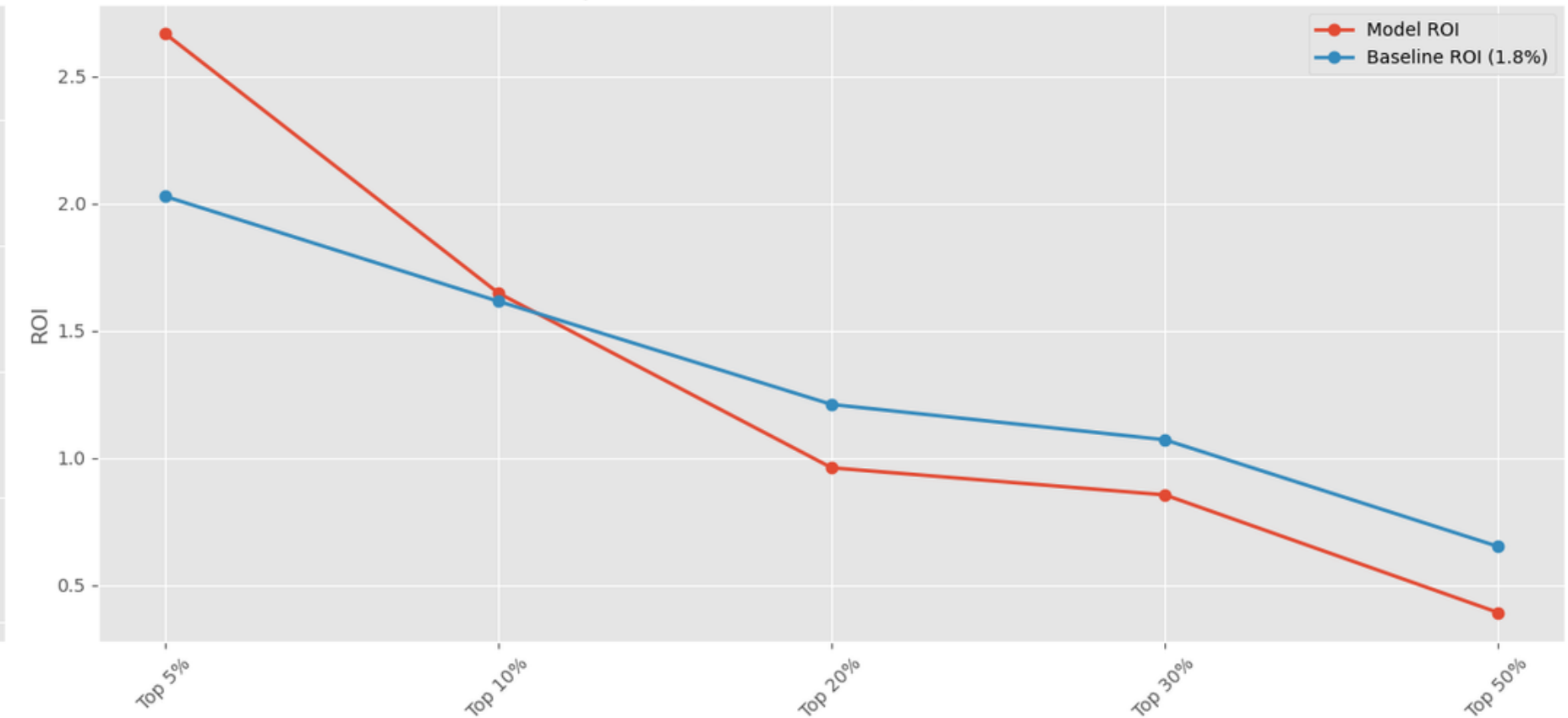
Profit Comparison (REAL Donations) — Model vs Baseline (1.8%)



€3380.8

Peak Profit at Top 10%

ROI Comparison (REAL Donations) — Model vs Baseline



2.65

ROI at Top 10%

WHY TARGET TOP 10%?

	Segment	Donors_Mailed	Responders_Any_Model	Responders_≥30_Model	Revenue_Model (€)	Cost_Model (€)	Profit_Model (€)	Expected_Responders_Baseline	Revenue_Baseline (€)	Profit_Baseline (€)
0	Top 5%	1282	59	44	3762.0	1025.6	2736.4	23	3106.0	2080.4
1	Top 10%	2564	87	67	5432.0	2051.2	3380.8	46	5366.0	3314.8
2	Top 20%	5129	133	96	8048.0	4103.2	3944.8	92	9070.0	4966.8
3	Top 30%	7693	187	136	11416.0	6154.4	5261.6	138	12750.0	6595.6
4	Top 50%	12822	237	174	14282.0	10257.6	4024.4	231	16966.0	6708.4

- The model predicts more responders than the 1.8% baseline.
- Even though the baseline appears to generate more revenue in a hypothetical scenario.
- It relies on knowing future donations – which is impossible.

WHY TARGET TOP 10%?

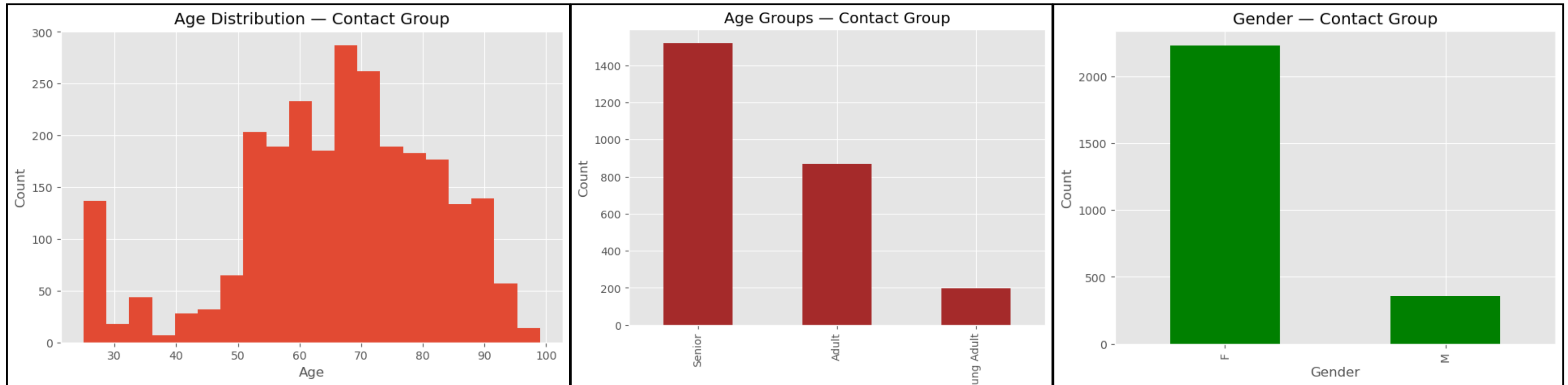
	Segment	Donors_Mailed	Responders_≥30	Responders_Any	Revenue (€)	Cost (€)	Profit (€)	ROI
0	Top 5%	1282	44	59	3762.0	1025.6	2736.4	3.668097
1	Top 10%	2564	67	87	5432.0	2051.2	3380.8	2.648206
2	Top 20%	5129	96	133	8048.0	4103.2	3944.8	1.961396
3	Top 30%	7693	136	187	11416.0	6154.4	5261.6	1.854933
4	Top 50%	12822	174	237	14282.0	10257.6	4024.4	1.392333
5	Bottom 10%	2564	1	25	224.0	2051.2	-1827.2	0.109204
6	Bottom 20%	5129	2	41	468.0	4103.2	-3635.2	0.114057
7	Bottom 30%	7693	14	61	1775.0	6154.4	-4379.4	0.288412
8	Bottom 50%	12822	43	124	4334.0	10257.6	-5923.6	0.422516

Highest ROI (2.65)

Strong profit (€3,381)

67 responders captured

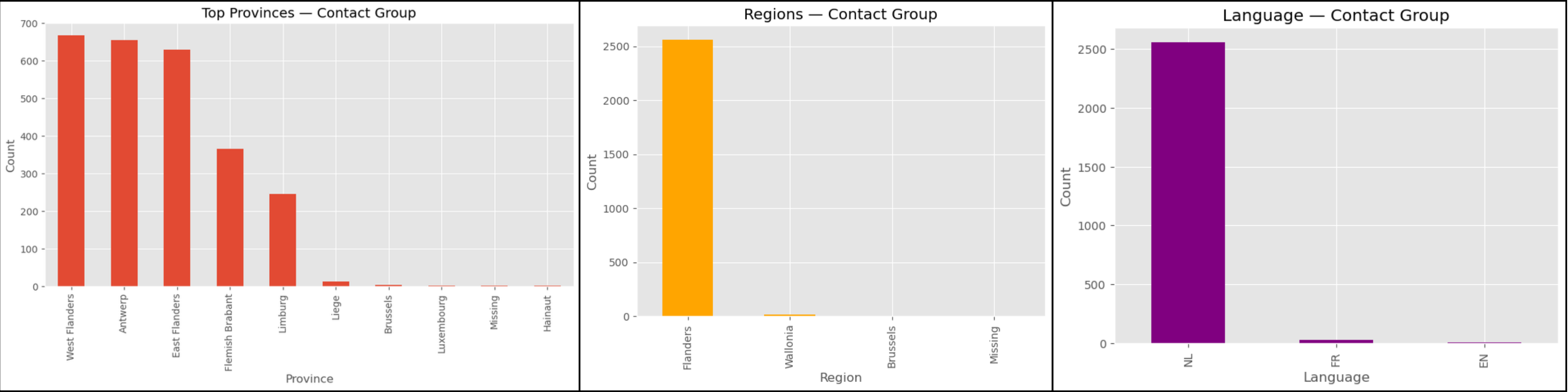
WHO ARE OUR TARGETED DONORS?



Seniors and Adults have higher probability to donate

Female donors → Higher donation likelihood than male donors

GEOGRAPHIC DISTRIBUTION



West Flanders ,Antwerp ,East Flanders show the highest number of potential responders

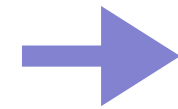
Dutch NL speakers → Higher predicted donation probability

KEY FINDINGS

- The model clearly outperforms random targeting.
- High predictive accuracy with a Test AUC in the high 0.80s.
- Behavioral donation patterns are the strongest predictors.
- The Top 10% of donors captures 35–40% of all responders.
- Only the Top 5–10% segments generate higher profit than the baseline.
- Mailing more than 20% becomes unprofitable when compared to baseline.

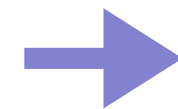
RECOMMENDATION

**Focus mailings on the Top
10%**



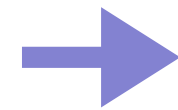
best ROI and highest response
rate

Avoid mailing beyond 20%



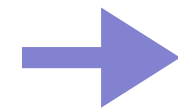
as additional letters do not generate enough revenue compared to
baseline

Adopt this model



as the targeting engine for top 10% population for all future
re-activation campaigns

Retrain periodically



to maintain performance as donor behaviour
evolves



Thank You